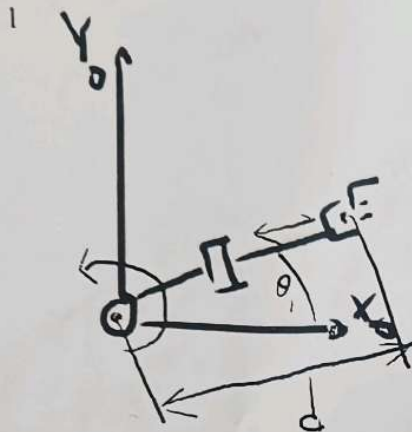




For the 1R-1P manipulator shown on the left, formulate the complete 6×2 Jacobian matrix. (6)
When $\theta = 60^\circ$ and $d = 200$ mm, find the linear and angular velocities of the end-effector E assuming the velocities of R and P joints to be 1 rad/s and 5 mm/s respectively.

2



For the 1R-1P manipulator in Q. 1, find the (5)
generalized joint torques required for pushing the surface with an end-point force of 50 N horizontally (in x_0 direction.) when $\theta = 30^\circ$ and $d = 100$ mm

- 3 In the expressions for actuator torques pertaining to a 2R planar manipulator, (4)
discuss the physical significance of terms involving square of joint velocity and product of the two joint velocities.